

Enhanced Motley–Keenan Physics for the 7th-Floor Path-Loss Simulator (**sim-v2.0**)

Indoor Walk Test v3 — physics specification

Abstract

This document specifies the electromagnetic model behind the v2 path-loss generator. It replaces v1’s fixed per-wall dB constants, global frequency multiplier, and empirical obstruction saturation with a physically grounded chain: complex permittivity from ITU-R P.2040-3; exact lossy-slab transmission that makes wall loss a function of incidence angle and thickness; a resistive-sheet model for the low-emissivity facade; Kouyoumjian–Pathak Uniform Theory of Diffraction (UTD) to fill shadow regions that v1 could only approximate with a saturation cap; and an explicit floor-attenuation term. Every equation below is implemented in `SIM/physics_v2.py` and `SIM/engine_v2.py` and verified by the self-tests in those files. The model predicts only path loss (rule R1); received-power metrics (RSRP/RSRQ/RSSI/SINR) are cheap arithmetic on top and are given in §7 for completeness. The one quantity all of this ultimately feeds is the surrogate’s training target, so the emphasis throughout is on quantities that are *measurable* and *identifiable* from the PCTEL G-flex scanner logs.

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1 Overview and notation

Let f be frequency (MHz unless a GHz subscript is shown), $\lambda_0 = c/f$ the free-space wavelength, and $k = 2\pi/\lambda_0$ the wavenumber. Incidence angles θ_i are measured from the surface normal. The time convention is $e^{+j\omega t}$, so a lossy medium carries a *negative* imaginary permittivity and the principal branch of $\sqrt{\cdot}$ is the decaying root throughout.

The total path loss along the direct ray is the Motley–Keenan form

$$L = L_{\text{FS}}(d, f) + \sum_i K_{w,i} L_{w,i}(f, \theta_i, t_i) + \sum_j K_{f,j} L_{f,j} + (\text{diffraction, combined in power}), \quad (1)$$

with free-space spreading using the exponent $n = 2$ (the walls, not the distance law, carry the environment):

$$L_{\text{FS}}(d, f) = 20 \log_{10} \left(\frac{4\pi d}{\lambda_0} \right), \quad d \geq d_0 = 1 \text{ m}. \quad (2)$$

Crossings $K_{w,i}$ count *contiguous runs* of material i along the ray (rule R3): a thick wall is one crossing, not one per raster cell. The novelty of v2 is that $L_{w,i}$ is a function of frequency, incidence angle and thickness rather than a constant; §2–5 give each factor.

2 Block A — Material electromagnetics (ITU-R P.2040-3)

Each material’s complex relative permittivity follows the P.2040-3 power laws (Table 3), evaluated with f in GHz:

$$\varepsilon'(f) = a f_{\text{GHz}}^b, \quad \sigma(f) = c f_{\text{GHz}}^d \text{ [S/m]}, \quad (3)$$

$$\varepsilon_r(f) = \varepsilon'(f) - j \frac{\sigma(f)}{2\pi f \varepsilon_0}. \quad (4)$$

Table 1 lists the coefficients actually used. The 619–627 MHz scanner bands sit below the 1 GHz floor of the recommendation’s validity range for most classes; the code flags this extrapolation rather than hiding it.

Table 1: P.2040-3 coefficients used, and the class each material backs.

Material	backs class	a	b	c	d
plasterboard	drywall partition	2.73	0	0.0085	0.9395
concrete	concrete / core	5.24	0	0.0462	0.7822
glass	exterior / interior glass	6.31	0	0.0036	1.3394
wood	furniture clutter	1.99	0	0.0047	1.0718

The transmitted angle follows complex Snell:

$$\sin \theta_t = \frac{\sin \theta_i}{\sqrt{\varepsilon_r}}. \quad (5)$$

Equation (5) matters more than it looks: for concrete ($\varepsilon' \approx 5.24$) a ray at 56° incidence refracts to only 21° inside the wall, so the internal path grows by 7%, not the 79% a naïve secant law would predict (§8).

3 Block B — Angle of incidence and wall thickness

Incidence angle and thickness are not two separate corrections; they are one slab problem, and the exact slab captures both.

3.1 Fresnel interface coefficients

For a wave in air striking the medium, the single-interface reflection coefficients for the two polarisations are

$$R_{\perp}(\theta_i) = \frac{\cos \theta_i - \sqrt{\varepsilon_r - \sin^2 \theta_i}}{\cos \theta_i + \sqrt{\varepsilon_r - \sin^2 \theta_i}} \quad (\text{TE}), \quad (6)$$

$$R_{\parallel}(\theta_i) = \frac{\varepsilon_r \cos \theta_i - \sqrt{\varepsilon_r - \sin^2 \theta_i}}{\varepsilon_r \cos \theta_i + \sqrt{\varepsilon_r - \sin^2 \theta_i}} \quad (\text{TM}). \quad (7)$$

3.2 Lossy-slab transmission

With electrical thickness

$$q(\theta_i, t, f) = \frac{2\pi t}{\lambda_0} \sqrt{\varepsilon_r - \sin^2 \theta_i}, \quad \text{Im } q < 0, \quad (8)$$

the coherent transmission through a slab of thickness t , summing the internal multiple reflections in amplitude, is

$$T(\theta_i, t, f) = \frac{(1 - R^2) e^{-jq}}{1 - R^2 e^{-2jq}}. \quad (9)$$

Equation (9) exhibits Fabry–Pérot ripple in $|T|$ whenever $q \approx m\pi$. That ripple is real but its *phase* is not knowable at raster resolution (one cell is 0.174 m, several wavelengths at 3.5 GHz), so the model uses a thickness-averaged power transmittance

$$\overline{|T|^2}(\theta_i, t, f) = \left\langle |T(\theta_i, t + \delta t, f)|^2 \right\rangle_{\delta t \sim \mathcal{N}(0, \sigma_t^2)}, \quad (10)$$

with $\sigma_t = \max(8 \text{ mm}, 0.1 t)$ capped at $t/2$. This is the honest form of “smooth over the ripple”: it is transparent in the thin limit ($t \ll \lambda$, where drywall lives at 3.5 GHz, so no loss is invented) and converges to the closed-form decohered transmittance in the thick limit. The per-crossing dielectric loss is then

$$L_w^{\text{diel}}(\theta_i, t, f) = -10 \log_{10} \overline{|T|^2}. \quad (11)$$

3.3 Sub-resolution metal (the fitted excess)

The 0.174 m raster cannot see the things that dominate measured wall loss above 2 GHz: steel studs inside partitions, rebar mesh in concrete, mullions and door frames in glazing, the metal doors of a lift core. These are metallic scatterers, so their contribution *grows* with frequency — exactly the residual left when the P.2040 dielectric slab is subtracted from measured per-wall loss. It is modelled as a fitted power law, scaled by a capped secant for oblique rays (an oblique ray crosses more studs):

$$E_w(\theta_i, f) = E_{\text{ref}} \left(\frac{f}{f_{\text{ref}}} \right)^{\gamma_e} \min \left(\frac{1}{\cos \theta_i}, 6 \right), \quad f_{\text{ref}} = 2442 \text{ MHz}. \quad (12)$$

Table 2: The seven material classes and their v2 parameters. t_{ref} is the nominal construction thickness (not the raster width); E_{ref}, γ_e are the fitted sub-resolution excess (12); R_s the low-E sheet resistance. Furniture accrues loss per metre of path, not per crossing.

id	class	P.2040	t_{ref} (m)	E_{ref} (dB)	γ_e	R_s ($\Omega/\square$)
0	air	—	—	—	—	—
1	drywall partition	plasterboard	0.026	2.38	1.20	—
2	concrete/masonry	concrete	0.200	1.71	1.46	—
3	core service area	concrete	0.200	8.06	0.36	—
4	furniture clutter	wood	1.000 [†]	—	—	—
5	exterior glass (low-E)	glass	0.006	—	—	25
6	interior glass	glass	0.010	1.60	1.28	—

[†] per-metre class; effective attenuation is the solid-wood value scaled by a fill fraction 0.021, reproducing the measured 0.30 dB m^{-1} at 2442 MHz.

It is identifiable separately from the slab because it has a different frequency signature, and the fitted values are physically legible (Table 2): a lift core comes out near 8 dB almost flat in f (metal doors), a stud partition near 2.4 dB rising as $f^{1.2}$. The total per-crossing loss is

$$L_w(\theta_i, t, f) = L_w^{\text{diel}}(\theta_i, t, f) + L_w^{\text{sheet}}(\theta_i, f) + E_w(\theta_i, f) \quad (13)$$

with the sheet term of §3.4 present only for the low-E facade.

3.4 Low-emissivity facade

The FCC HQ facade is metal-coated low-E glass. A thin conductive film is a shunt sheet admittance $1/R_s$ on a transmission line of wave impedance $\eta_{\text{TE}} = \eta_0 / \cos \theta_i$ or $\eta_{\text{TM}} = \eta_0 \cos \theta_i$, giving power transmittance

$$T^{\text{sheet}}(\theta_i) = \left(\frac{2}{2 + \eta/R_s} \right)^2, \quad L_w^{\text{sheet}} = -10 \log_{10} T^{\text{sheet}}, \quad (14)$$

cascaded incoherently with the dielectric slab. A sheet resistance $R_s = 25 \Omega \text{ sq}^{-1}$ — squarely in the $4 \Omega \text{ sq}^{-1}$ to $40 \Omega \text{ sq}^{-1}$ range of real low-E coatings — reproduces the measured 20 dB at normal incidence. The consequence is the model’s headline result: because η_{TE} grows with angle, the same facade costs 34 dB at 75° , so a street-level macro striking the glass obliquely penetrates far less than v1’s fixed 15 dB O2I assumption — directly strengthening the co-channel interference argument.

4 Block C — Diffraction (UTD)

v1 sent deep-shadow loss to +200 dB along straight rays and capped it with an empirical saturation. v2 instead supplies the energy that really arrives around corners, using the Kouyoumjian–Pathak coefficient for a wedge of exterior angle $n\pi$:

$$D_{s,h} = \frac{-e^{-j\pi/4}}{2n\sqrt{2\pi k} \sin \beta_0} \left[\cot\left(\frac{\pi+(\phi-\phi')}{2n}\right) F(kLa^+(\phi-\phi')) + \cot\left(\frac{\pi-(\phi-\phi')}{2n}\right) F(kLa^-(\phi-\phi')) \right. \\ \left. \pm \left(\cot\left(\frac{\pi+(\phi+\phi')}{2n}\right) F(kLa^+(\phi+\phi')) + \cot\left(\frac{\pi-(\phi+\phi')}{2n}\right) F(kLa^-(\phi+\phi')) \right) \right], \quad (15)$$

with the distance parameter $L = ss'/(s + s')$ for spherical incidence, the sign $-$ for the soft (D_s) and $+$ for the hard (D_h) coefficient, and

$$a^\pm(\beta) = 2 \cos^2\left(\frac{2\pi n N^\pm - \beta}{2}\right), \quad N^\pm = \text{nearest integer to } \frac{\beta \pm \pi}{2\pi n}. \quad (16)$$

The transition function is

$$F(X) = 2j\sqrt{X} e^{jX} \int_{\sqrt{X}}^{\infty} e^{-j\tau^2} d\tau, \quad F(X) \rightarrow 1 \ (X \rightarrow \infty), \quad F(X) \rightarrow \sqrt{\pi X} e^{j(\pi/4+X)} \ (X \rightarrow 0), \quad (17)$$

evaluated through the Fresnel integrals. The diffracted-ray path loss, referenced identically to L_{FS} so it is directly comparable with the direct ray, is

$$L_{\text{diff}} = 20 \log_{10}\left(\frac{4\pi}{\lambda_0}\right) + 10 \log_{10}(s's(s + s')) - 20 \log_{10}|D_{s,h}|, \quad (18)$$

which folds in the spherical spreading factor $A(s, s') = \sqrt{s'/(s(s + s'))}$.

Lossy wedges (Luebbers). Building corners are not perfect conductors. The reflection-boundary terms are weighted by the Fresnel coefficient of the reflecting face, replacing the PEC weights ∓ 1 with the face's $R(\theta_i)$; this couples Block C back to Block B rather than leaving diffraction as a bolt-on.

Why this matters and what it replaces. The coefficient scales as $D \propto \sqrt{\lambda}$, so diffracted loss grows as $30 \log_{10} f$ against free space's $20 \log_{10} f$. This is precisely the relation $S = \gamma\sqrt{\lambda} = (\gamma_D\sqrt{\lambda}) C_e = D \cdot C_e$: a diffraction coefficient times a per-situation correction. Recursion is capped at 2–3 reflections plus 1–2 diffractions; beyond that the contributions fall below the noise floor. With Block C in place, **the v1 obstruction saturation constant is removed** — the loss self-limits for the right reason, and the fraction of indoor cells above the clip ceiling drops rather than being pinned there artificially.

5 Block D — Floor attenuation

For n_f floor crossings the model uses the physical form — each floor is a Block B slab with a rebar excess —

$$L_f(n_f) = n_f \left[L_w^{\text{diel}}(\theta_i, t_{\text{floor}}, f) + E^{\text{rebar}} \right], \quad (19)$$

linear in n_f . The empirical alternative $L_f = 15 + 4(n_f - 1)$ dB saturates, but that saturation is the signature of the dominant path switching from straight-through to out-the-facade-and-back-in for $n_f \geq 3$ — a mechanism Block C represents explicitly, so (19) is preferred and the two are never stacked. On the single deployed floor this term is scaffolded, not exercised.

6 Field combination

Direct and diffracted contributions combine in *power* (rule R7),

$$P_{\text{tot}} = \sum_p |E_p|^2 \iff 10^{-L_{\text{tot}}/10} = \sum_p 10^{-L_p/10}, \quad (20)$$

not in amplitude. RSRP is already averaged over resource elements and the H3 receiver bins average over metres of motion, so a coherent sum would inject fast fading that cannot be calibrated against and would only inflate the residuals. The result is floored at free space — a passive obstructed channel cannot beat free space — which is exactly the D.3 sanity check the pipeline runs.

7 Metric mapping

The surrogate predicts path loss only; the received-power metrics are arithmetic on top, applied downstream so no transmit parameter enters the learned model (rule R2):

$$\text{RSRP} = P_{\text{tx}} - 10 \log_{10}(12N_{\text{RB}}) + G_{\text{tx}} + G_{\text{rx}} - L - L_{\text{cable}}, \quad (21)$$

$$\text{RSSI} = 10 \log_{10} \left(\sum_{\text{cells}} 10^{P_{\text{rx}}/10} + 10^{N_0 B/10} \right), \quad (22)$$

$$\text{RSRQ} = 10 \log_{10}(N_{\text{RB}}) + \text{RSRP} - \text{RSSI}, \quad (23)$$

$$\text{SINR} = \frac{P_s}{\sum_i P_i + N_0 B}, \quad N_0 B = -174 + 10 \log_{10} B + \text{NF} \text{ [dBm]}. \quad (24)$$

8 Calibration (Phase D)

Every material constant above is literature until fitted to this building. With a known-transmitter walk, the free parameters

$$\theta = \{ E_{\text{ref},m}, \gamma_{e,m}, R_s, t_m, P_{\text{tx}}, L_{\text{cable}} \} \quad (25)$$

are fit by penalised least squares against measured RSRP,

$$\mathcal{L}(\theta) = \sum_k w_k (\text{RSRP}_k^{\text{pred}} - \text{RSRP}_k^{\text{meas}})^2 + \lambda \|\theta - \theta_{\text{P.2040}}\|^2, \quad (26)$$

with the prior term keeping the fitted permittivities physical. The multi-band scanner logs identify γ_e per material (the same wall seen at 619/1935/2600 MHz gives three points); 20 % of measured points are held out as the honest field-test set, with acceptance ≤ 8 dB RMSE, the shadow-fading floor. The `jitter` field stored with every dataset sample is what makes recalibration a fine-tune rather than a full retrain.

Implementation and verification

Blocks A–D are in `SIM/physics_v2.py` (with ~ 30 physics self-tests: permittivity sign and dispersion, Fresnel limits and passivity, slab thin/thick limits and the incoherent-average identity, the low-E 20 dB anchor, the UTD half-field normalisation and $\sqrt{\lambda}$ scaling, and the material table against the measured anchors). The raster geometry — crossing extraction, incidence angle, thickness, edge detection and the diffraction fill — is in `SIM/engine_v2.py`, with reciprocity and no-saturation tests. A CUDA port, `SIM/engine_v2_torch.py`, is verified to match the NumPy reference to a path-loss $p_{99.99}$ under 0.05 dB over the training-relevant range.

A note on the angle law

The common “enhanced MK” recipe scales wall loss by $\sec \theta_i$. For high-permittivity building materials this over-attenuates oblique rays substantially, because Snell (5) refracts the ray toward the normal inside the wall: the absorbing path grows by $1/\cos \theta_t \ll \sec \theta_i$, and the extra loss at oblique incidence is mostly increased Fresnel *reflection*, which the exact slab (10) already captures. At 56° on concrete the exact slab gives $1.26\times$ the normal-incidence loss where the secant law would claim $1.79\times$; at 80° the gap is $2.1\times$ versus $5.8\times$. Using the exact slab is therefore both more correct and less prone to inventing loss that the scanner would not see.